SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I P. HEMANTH REDDY (192312173) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ____→ Transport → Network → ____→ Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Ans:

Missing Layers

Application → Presentation → Session → Transport → Network → Data Link → Physical.

Session layer establishes, manages and terminates communication sessions. Data Link layer provides framing, MAC addressing, error detection and reliable node-to-node delivery. Together they ensure organized communication and reliable transfer.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Ans:

Application/Presentation/Session: Data

Transport: Segment

Network: Packet

Data Link: Frame

Physical: Bits

As data moves downward, each layer adds its own header (encapsulation): Data→Segment→Packet→Frame→Bits. At the receiver, decapsulation removes headers in reverse order, ensuring correct delivery and interpretation.

3. Given the concept map:

DataLink→MACAddress→LocalDeli

very Network → IP Address → Global

Delivery

Explain how these relationships ensure complete data delivery across networks.

Ans:
Data Link → MAC Address → Local Delivery
Network → IP Address → Global Delivery
MAC addresses deliver frames within the same local network, while IP addresses route packets across different networks. Routers use IP for end-to-end routing, and each local hop uses MAC addresses, enabling complete delivery.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Ans:

Error Control

├─ Data Link Layer → Error Detection

└─ Transport Layer → Error Correction/Recovery

The Data Link layer detects transmission errors on each link (e.g., CRC). The Transport layer (e.g., TCP) recovers from lost/corrupted segments using acknowledgements, sequencing and retransmission. Together they improve reliability.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Ans:
Application (Browser/HTTP) → Presentation (Encryption/TLS, formatting) → Session (Session management) → Transport (TCP) → Network (IP routing) → Data Link (Frames, MAC) → Physical (Bits over cable/Wi-Fi).

Failure at any layer affects higher layers. For example, if the Physical layer fails, no bits are transmitted; if Transport fails, webpages may not load completely despite a working network.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

Ans:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

The physical connection and local communication are working, but the Network layer has failed (such as incorrect IP address, routing problem, or router failure). Since Transport depends on Network, and Application depends on Transport, all upper layers also fail. The concept map helps isolate the first failing layer, making troubleshooting faster.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation